

Gauss Law, Monodromy Defect, Magnetic Lattice Translation, and Lattice Duality

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Abstract

In a Hamiltonian lattice gauge theory, the physical Hilbert space is fixed by the Gauss law, by choosing an eigenspace of the Gauss law operator which generates the gauge transformation. Choosing a non-trivial eigenspace is referred to as the modified Gauss law. In 2+1d dynamical U(1) pure gauge theory, we show that these physical Hilbert spaces are defect Hilbert spaces associated with non-topological monodromy defects for monopole operators and topological defects for U(1) 1-form symmetry, inserted along the time direction. We also construct the corresponding lattice translation operators. Under a nontrivially modified Gauss law, these translations obey a magnetic translation algebra. Finally, we discuss how the modified Gauss laws are manifested in 2+1d compact bosons, via exact duality on the lattice.

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1 Introduction

In a quantum field theory or lattice model with flat or dynamical gauge fields, physical observables must be gauge invariant. In a Hamiltonian lattice model, one typically begins with an extended tensor-product Hilbert space $\mathcal{H}$. Gauge transformations acting on $\mathcal{H}$ are generated by the Gauss law operator $G[\zeta(x)]$, where $\zeta(x)$ is a spatially dependent gauge parameter. The physical Hilbert space is defined as the subspace of $\mathcal{H}$ satisfying

$$G[\zeta(x)] = 1 \quad \text{for all } \zeta(x). \quad (1.1)$$

The constraint (1.1) is referred to as the Gauss law.

In fact, the notion of Gauss laws is suggested to be generalized to any constraint imposed on the Hilbert space, not necessarily associated with the gauge group as an analog from the continuum. See more discussions in footnote 3.

However, we are not strictly required to define the physical Hilbert space as the eigenspace of $G[\zeta(x)]$ with eigenvalue 1. Instead, we can choose a more general eigenspace governed by a modified Gauss law:

$$G[\zeta(x)] = \chi[\zeta(x)], \quad (1.2)$$

where χ takes values in $U(1)$ since G is unitary. The modified Gauss law (1.2) defines a distinct physical Hilbert space, denoted by $\mathcal{H}_\chi$.

A natural question then arises: *what is the interpretation of $\mathcal{H}_\chi$ for a nontrivial $\chi \neq 1$, and how are the spaces $\mathcal{H}_\chi$'s related for inequivalent χ 's?*

For flat gauge fields of finite groups, this question has been studied in the context of discrete gauging [1–5]. The answer is that the modification of Gauss law at one site¹ corresponds to inserting a Wilson line along the time direction at this site. The Wilson line is topological since the gauge field is flat.

The answer can be seen as follows. For a flat 1-form gauge field, we only need a pair conjugate variables on each links. If we denote the local matter charge as Q_s and the conjugate variable to A_ℓ as E_ℓ , then the Gauss law operator is

$$G(\zeta) = \exp \left(i \sum_s \zeta_s (Q + \delta E)_s \right), \quad (1.3)$$

for ζ_s being an element in the finite gauge group. See Appendix A for the definition of δ . The Wilson lines generate a $(D - 2)$ -form symmetry² [6–11], where D is the spacetime dimension. Inserting a Wilson line $\exp(ik \int A)$ at site s_* is done by conjugating the system (i.e. Gauss law and Hamiltonian) by the unitary

$$\exp \left(ik \sum_{\ell \in L} A_\ell \right), \quad (1.4)$$

where L is a half-line ending on s_* . This shift G by a phase only depends on ζ_{s_*} ,

$$G(\zeta) \rightarrow e^{ik\zeta_{s_*}} G(\zeta). \quad (1.5)$$

¹Here we consider 1-form gauge field. The stories are the same for higher form gauge fields.

²i.e. the quantum symmetry in the context of discrete gauging.

It can be checked that starting with $\mathcal{H}_1$ defined by (1.1), one can obtain $\mathcal{H}_\chi$ for any nontrivial χ by inserting appropriate Wilson lines at each site. Equivalently, the physical Hilbert space for a modified Gauss law is the defect Hilbert space, twisted by some Wilson lines.

For a dynamical U(1) gauge field [1, 5, 12–15], because of the non-flat field strength, we can have a monopole configuration. To realize the monopole operator and the monopole number operator on the lattice, we need to introduce the Villain field N , which is a 2-form $\mathbb{Z}$ gauge field. One way to think about this is that we start with a $\mathbb{R}$ gauge field and then flat gauge the $\mathbb{Z}^{(1)}$ symmetry [16–26]. Or we just view N as one of the differential cohomology defining data for a U(1) gauge field in its own right [27–31]. See [31] for a pedagogical introduction, and the N and A in our notation is almost the discrete version of theirs.

Now, with more gauge parameters and more subtle gauge transformations, the Gauss law operator takes a more complicated form than (1.3). Without the previous $(D-2)$ -form symmetry since the Wilson lines are *not* topological, one might expect the $U(1)^{(D-3)}$ magnetic symmetry defect will save the day. But as we show in Section 3, these topological defect insertions do *not* generate all the modified Gauss laws, i.e. nontrivial χ 's.

Main results In this work we focus on U(1) pure gauge theory in $2+1$ d. We show that the consistent modified Gauss laws are classified by real lattice 1-cochain $\lambda \in C^1(\Lambda, \mathbb{R})$ modulo closed $2\pi\mathbb{Z}$ -valued 1-cochains and a 2π periodic 2-cochain $\theta \in C^2(\Lambda, \mathbb{R}/2\pi\mathbb{Z})$. More explicitly, the modified local Gauss laws are

$$(\delta E)_s = \frac{(\delta\lambda)_s}{2\pi}, \quad e^{i(2\pi E + \delta B)_\ell} = e^{i\lambda_\ell}, \quad e^{2\pi i N_p} = e^{i\theta_p} \quad (1.6)$$

We then identify the physical meaning of these sectors. A non-zero λ corresponds to inserting *open* magnetic surface defects

$$\exp\left(i\lambda \int_\Sigma \frac{F}{2\pi}\right), \quad \partial\Sigma \neq 0. \quad (1.7)$$

The boundary of such an open surface is a monodromy defect for monopole operators, also referred to as the twist defect. On the lattice, the monodromy at a site s is $e^{i(\delta\lambda)_s}$. And a non-zero θ corresponds to electric $U(1)_e^{(1)}$ topological defects, or a vortex defect when the electric $U(1)_e^{(1)}$ symmetry is explicitly broken by some matter fields.

Then, we construct the translation operators acting on these defect Hilbert spaces $\mathcal{H}_\chi$.

These translations obey

$$T_1 T_2 T_1^{-1} T_2^{-1} = \exp \left[-i \sum_s (\delta\lambda)_s \left(N + \frac{dA}{2\pi} \right)_{s+\frac{\hat{x}}{2}+\frac{\hat{y}}{2}} \right], \quad (1.8)$$

which is a magnetic-translation algebra. The noncommutativity of translations is therefore the manifestation of the monodromy defects.

Finally we demonstrate the exact duality to compact boson as a Hamiltonian lattice model, and compare the modified Gauss law on both sides.

The paper is organized as follows. In Section 2, we explain the modified Gauss law in dynamical U(1) gauge theory, and work out the moduli space of physical Hilbert spaces. Section 3 shows that a non-trivially modified Gauss law describes the Hilbert space with dynamical monodromy defect insertions. In Section 4, the translation operation on each Hilbert space is constructed, with the symmetry algebra agreeing with the monodromy. Section 5 is about the exact lattice model duality and some discussions about matter are made in Section 6.

Our conventions and notations are collected in Appendix A, a proof is given in Appendix B, and the comparison with the Hamiltonian lattice gauge theory of a finite group is made in Appendix C.

2 2+1d dynamical U(1) gauge theory

2.1 Modified Gauss law

We explain the modification of Gauss law in a Hamiltonian lattice model, with only U(1) gauge field, and without matter degrees of freedom.

The spatial lattice is a 2d square lattice with periodic boundary conditions along both x and y directions. We denote Λ for the lattice, s for sites, ℓ for links and p for plaquettes.

We recall the Hilbert space for dynamical U(1) gauge fields on the lattice as follows. The idea follows from Villain [17] where one represents a U(1) gauge field by an $\mathbb{R}$ gauge field and a $\mathbb{Z}$ gauge field, see also recent discussions [18, 22, 13, 23, 19, 20]. On each link we place a pair of $\mathbb{R}$ -valued conjugate variables,

$$[A_\ell, E_{\ell'}] = i\delta_{\ell\ell'}, \quad (2.1)$$

and the local Hilbert space on each link forms a representation of this algebra $\mathcal{H}_\ell \simeq L^2(\mathbb{R})$.

On each plaquette we place a pair of $\mathbb{R}$ -valued conjugate variables,

$$[B_p, N_{p'}] = i\delta_{pp'}, \quad (2.2)$$

and the local Hilbert space on each plaquette forms a representation of this algebra $\mathcal{H}_p \simeq L^2(\mathbb{R})$. We refer to the tensor product Hilbert space $\mathcal{H} = \otimes_p \mathcal{H}_p \otimes_\ell \mathcal{H}_\ell$ as the extended Hilbert space.

The tensor product Hilbert space is subjected to two types of Gauss laws. The first Gauss law enforces the eigenvalue of N_p to be integers by demanding

$$\exp(2\pi i \tilde{n}_p N_p) \quad (2.3)$$

to be 1 for every plaquette, for an arbitrary

$$\tilde{n} \in C^2(\Lambda, \mathbb{Z}). \quad (2.4)$$

Later, we allow more general Gauss laws by allowing (2.3) to be an arbitrary constant phase, not necessarily 1. This is why we start by setting N_p to be $\mathbb{R}$ valued instead of $\mathbb{Z}$ valued. The Gauss law of N_p also generates a gauge transformation of its conjugate variable B_p

$$B_p \rightarrow B_p + 2\pi \tilde{n}_p. \quad (2.5)$$

This Gauss law does not break the tensor product structure of $\mathcal{H}$.

To see the second Gauss law, it is more natural to start with the gauge transformation. The $U(1)$ gauge field is represented by a pair (N_p, A_ℓ) , with the gauge transformation

$$\begin{aligned} A_\ell &\rightarrow A_\ell + 2\pi n_\ell + (d\alpha)_\ell, \\ N_p &\rightarrow N_p - (dn)_p, \end{aligned} \quad (2.6)$$

where

$$n \in C^1(\Lambda, \mathbb{Z}), \quad \alpha \in C^0(\Lambda, \mathbb{R}). \quad (2.7)$$

More concretely, n_ℓ 's are $\mathbb{Z}$ -valued numbers on links and α_s 's are $\mathbb{R}$ -valued numbers on sites. See Appendix A for the conventions for (co)chains, operator d and its adjoint δ . The gauge transformation (2.6) is generated by two Gauss law operators

$$\exp\left(i \sum_\ell n_\ell (2\pi E_\ell + (\delta B)_\ell)\right), \quad \exp\left(i \sum_s \alpha_s (\delta E)_s\right) \quad (2.8)$$

The first one generates the $\mathbb{Z}$ redundancy of A_ℓ , which effectively makes it compact. The second one is the standard Gauss law for $U(1)$.³ This Gauss law breaks the tensor product structure of $\mathcal{H}$.

One can package the Gauss laws (2.3) and (2.8) into a single one⁴

$$G(n, \alpha, \tilde{n}) = \exp \left(i \sum_{\ell} n_{\ell} (2\pi E_{\ell} + (\delta B)_{\ell}) + i \sum_s \alpha_s (\delta E)_s \right) \exp \left(2\pi i \sum_p \tilde{n}_p N_p \right). \quad (2.9)$$

The operator $\delta : C^k(\Lambda) \rightarrow C^{k-1}(\Lambda)$ can be thought as the lattice analogy for divergence and is defined in Appendix A. Note that instead of writing the Gauss law operator as a local operator, say at one site (for one non-zero α_s), we defined it as the operator implementing any gauge transformation parameterized by $(n, \alpha, \tilde{n}) \in C^1(\Lambda, \mathbb{Z}) \times C^0(\Lambda, \mathbb{R}) \times C^2(\Lambda, \mathbb{Z})$. This significantly simplifies the discussions below.

The physical Hilbert space is a subspace $\mathcal{H}_{\chi}$ of the tensor product Hilbert space $\mathcal{H}$. More concretely, $\mathcal{H}_{\chi}$ is an eigenspace of the Gauss law $G(n, \alpha, \tilde{n})$ with eigenvalue $\chi(n, \alpha, \tilde{n}) \in U(1)$, i.e.

$$\mathcal{H}_{\chi} = \{|\psi\rangle \in \mathcal{H} : G|\psi\rangle = \chi|\psi\rangle\} \subset \mathcal{H}. \quad (2.10)$$

Traditionally [1, 13], χ is taken to be 1 hence $G(n, \alpha, \tilde{n}) = 1$, reducing to

$$e^{i(2\pi E_{\ell} + (\delta B)_{\ell})} = 1, \quad (\delta E)_s = 0, \quad e^{2\pi i N_p} = 1. \quad (2.11)$$

(2.11) is usually referred to as *the* Gauss law for $U(1)$ gauge theory on the lattice. But generally, χ can be a $U(1)$ -valued function

$$\chi : C^1(\Lambda, \mathbb{Z}) \times C^0(\Lambda, \mathbb{R}) \times C^2(\Lambda, \mathbb{Z}) \rightarrow U(1). \quad (2.12)$$

We refer to $G = \chi$ for nontrivial χ as a *modified Gauss law*.

We make some remarks on χ .

1. Firstly, different terms in (2.9) commute. In particular, the $\exp(in_{\ell}(\delta B)_{\ell})$ from the first term commutes with $\exp(2\pi i \tilde{n}_p N_p)$ from the second term. It follows that χ is a group

³In the continuum we typically refer to the transformation associated with the gauge group as the gauge transformation, which is (2.6) in this case. And we call (2.5) the identification in the definition of a periodic scalar, rather than treating it as a Gauss law. However, on the lattice both these are realized as constraints, or the Gauss laws, so they should not be view as different kinds. In fact, in Section 5, we see this two roles get exchanged under the duality.

⁴The term exponents can not be placed in a single exponent because $(\delta B)_{\ell}$ does not commute with N_p when $\ell \in \partial p$. But the two exponents commute.

homomorphism between abelian groups, i.e. $\chi(n_1 + n_2, \alpha, \tilde{n}) = \chi(n_1, \alpha, \tilde{n})\chi(n_2, \alpha, \tilde{n})$, etc.

2. Secondly, χ satisfies some consistency conditions, such as $\chi(dk, -2\pi k, 0) = G(dk, -2\pi k, 0) = 1$ by the definition of G (2.9). We will work out the space of consistent χ 's (2.19), that is, the moduli space physical Hilbert space $\mathcal{H}_\chi$. Each point in the moduli space fixes a physical Hilbert space on the lattice.
3. Thirdly, we only specify the gauge field degrees of freedom. The summation over site terms in G (2.9) should include matter local charges if we intend to include the matter. Then the moduli space of χ will be different. Here we focus on pure gauge theory. Including matter is briefly discussed in Section 6.

2.2 Moduli space of physical Hilbert spaces

We work out a way to characterize different χ 's, hence obtaining the moduli space for physical Hilbert spaces $\mathcal{H}_\chi$.

Being a group homomorphism, $\chi(n, \alpha, \tilde{n}) = \chi(n, 0, 0)\chi(0, \alpha, 0)\chi(0, 0, \tilde{n})$, $\chi(0, 0, \tilde{n})$ can be easily fixed, because N_p only appears in the third factor of G . But the variable E_ℓ is involved in the first two factors of G simultaneously, so $\chi(n, \alpha, 0)$ needs to be treated with more care.

We start by looking at $\chi(0, 0, -) : C^2(\Lambda, \mathbb{Z}) \rightarrow \text{U}(1)$, since any one dimensional representation of $\mathbb{Z}$ is labeled by a $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, i.e. $\rho_\theta(n) = e^{i\theta n}$, there exists a unique set of $\{\theta_p \in \mathbb{R}/2\pi\mathbb{Z}\}$, i.e. $\theta \in C^2(\Lambda, \mathbb{R}/2\pi\mathbb{Z})$, such that

$$\chi(0, 0, \tilde{n}) = \exp \left(i \sum_p \tilde{n}_p \theta_p \right). \quad (2.13)$$

Then, for $\chi(0, -, 0) : C^0(\Lambda, \mathbb{R}) \rightarrow \text{U}(1)$, since any one dimensional representation of $\mathbb{R}$ is labeled by a real number $q \in \mathbb{R}$, i.e. $\rho_q(x) = e^{iqx}$, there exists a unique set of $\{q_s \in \mathbb{R}\}$, i.e. $q \in C^0(\Lambda, \mathbb{R})$, such that

$$\chi(0, \alpha, 0) = \exp \left(i \sum_s \alpha_s q_s \right). \quad (2.14)$$

Lastly, for $\chi(-, 0, 0) : C^1(\Lambda, \mathbb{Z}) \rightarrow \text{U}(1)$, there exists a unique set of $\{\varphi_\ell \in \mathbb{R}/2\pi\mathbb{Z}\}$, i.e.

$\varphi \in C^1(\Lambda, \mathbb{R}/2\pi\mathbb{Z})$, such that

$$\chi(n, 0, 0) = \exp \left(i \sum_{\ell} n_{\ell} \varphi_{\ell} \right). \quad (2.15)$$

We claim that there exist $\lambda \in C^1(\Lambda, \mathbb{R})$, such that

$$q_s = \frac{(\delta\lambda)_s}{2\pi} \in \mathbb{R}, \quad \text{and} \quad \varphi_{\ell} = \lambda_{\ell} \mod 2\pi. \quad (2.16)$$

This is proven in Appendix B, where λ is constructed, using some consistency conditions in χ mentioned in the above remarks. This is not an obvious fact which can easily be seen by modifying the right-hand side of the traditional local Gauss law (2.11). Thus, the function χ can be written as

$$\chi(n, \alpha, \tilde{n}) = \exp \left(i \sum_{\ell} n_{\ell} \lambda_{\ell} + i \sum_s \alpha_s \frac{(\delta\lambda)_s}{2\pi} + i \sum_p \tilde{n}_p \theta_p \right). \quad (2.17)$$

Thus, on the physical Hilbert space

$$(\delta E)_s = \frac{(\delta\lambda)_s}{2\pi}, \quad e^{i(2\pi E + \delta B)_{\ell}} = e^{i\lambda_{\ell}}, \quad e^{2\pi i N_p} = e^{i\theta_p} \quad (2.18)$$

Note that λ and λ' determine the same χ by (2.17) if and only if $\lambda' - \lambda \in C^1(\Lambda, 2\pi\mathbb{Z})$ and $\delta\lambda = \delta\lambda'$. Thus, the space of inequivalent λ and θ , is

$$\mathcal{M} = \frac{C^1(\Lambda, \mathbb{R})}{C^1(\Lambda, 2\pi\mathbb{Z}) \cap \ker \delta} \times C^2(\Lambda, \mathbb{R}/2\pi\mathbb{Z}). \quad (2.19)$$

Any representative element $(\lambda, \theta) \in \mathcal{M}$ fixes a χ by (2.17)⁵, hence by (2.10) fixing a physical Hilbert space $\mathcal{H}_{\chi}$ of pure U(1) gauge theory. We say that $\mathcal{M}$ is the *moduli space of physical Hilbert spaces*.

2.3 Different presentations

Our discussion suffices for any Hamiltonian lattice model with constraints, not necessarily those coming from the obvious analog of the gauge theory in the continuum, one example will be given in Section 5. So what we mean by the modified Gauss laws are actually the modified constraints on physical Hilbert spaces. We use *the Gauss laws* and *the constraints*

⁵We identify $\mathcal{M}$ with the space of χ 's, and just write $\chi \in \mathcal{M}$ hoping without confusion.

interchangeably.

The lattice model is defined by the pair $(G = \chi, H)$, where H is the Hamiltonian. It is known [5, 12] that we can perform some unitary transformation to the system, shifting between different presentations of the theory, for example from the modified Gauss laws to the modified Hamiltonian ($G' = UGU^{-1} = 1, H' = UHU^{-1}$).

Note that the unitary U is the operator on the total Hilbert space $\mathcal{H}$, not just the unitary operators on physical space $\mathcal{H}_\chi$. Although the unitary transformation does not change the spectrum of H in $\mathcal{H}$, the spectrum of $H|_{\mathcal{H}_\chi}$ is changed since the physical Hilbert space is changed. We will see such examples below. This is consistent with the fact that modification of the constraints changes the energy spectrum.

Consider the following standard Maxwell theory Hamiltonian on the lattice,

$$H = \sum_p \frac{1}{2e^2} (2\pi N + dA)_p^2 + \sum_\ell \frac{e^2}{2} E_\ell^2, \quad (2.20)$$

with the *modified* constraints (2.18), where the two terms are the energy for magnetic and electric fields, respectively. We can perform the unitary to do the shift,

$$U : \quad E \rightarrow E + \frac{\lambda}{2\pi}, \quad N \rightarrow N + \frac{\theta}{2\pi}, \quad (2.21)$$

which is realized by $U = \exp\left(-\frac{i}{2\pi}\langle\lambda, A\rangle - \frac{i}{2\pi}\langle\theta, B\rangle\right)$ (see (A.8) for the shorthand notation of inner product). In this new presentation, the Gauss laws are unmodified as (2.11), but the Hamiltonian gets modified

$$H' = \sum_p \frac{1}{2e^2} (2\pi N + dA + \theta)_p^2 + \sum_\ell \frac{e^2}{2} \left(E + \frac{\lambda}{2\pi}\right)_\ell^2. \quad (2.22)$$

See the summary in Table 1. Of course, we can introduce some intermediate presentation, by conjugating the unitary which only shift one of N and E , where both the Gauss law and Hamiltonian get modified. We will use different presentations interchangeably, since they defined the same theory.

Note that since we have classified all possible modified Gauss laws in Section 2.2, which is independent of the Hamiltonian, the unitary transformation (2.21) to another presentation works for any Hamiltonian that commutes with the Gauss law.

Presentations	Modified Gauss law, Unmodified Hamiltonian	Unmodified Gauss law, Modified Hamiltonian
Gauss laws	$\delta E = \frac{\delta \lambda}{2\pi}$ $e^{i(2\pi E + \delta B)} = e^{i\lambda}$ $e^{2\pi i N} = e^{i\theta}$	$\delta E = 0$ $e^{i(2\pi E + \delta B)} = 1$ $e^{2\pi i N} = 1$
Hamiltonian	$H = \sum_p \frac{1}{2e^2} (2\pi N + dA)_p^2$ $+ \sum_\ell \frac{e^2}{2} E_\ell^2$	$H' = \sum_p \frac{1}{2e^2} (2\pi N + dA + \theta)_p^2$ $+ \sum_\ell \frac{e^2}{2} \left(E + \frac{\lambda}{2\pi} \right)_\ell^2$

Table 1: Two presentations of the modified Maxwell theory on the lattice. The transformation between the two presentations is achieved by the unitary (2.21).

3 Moduli space and defects

To understand the meaning of nontrivial χ 's, we need to take a digression and discuss inserting (dynamical) defect in the Hamiltonian lattice model. Here defects are to be contrasted with operators, where the former stretch in the time direction.

We show that for a nontrivial χ (2.17), determined by $(\lambda, \theta) \in \mathcal{M}$, the non-zero θ corresponds to the insertion of topological $U(1)_e^{(1)}$ line defects, and the non-zero λ corresponds to the insertion of *non-topological* monodromy (or twist) defects for $U(1)_m^{(0)}$ open surface defect. Hence the physical Hilbert space $\mathcal{H}_\chi$ is the corresponding defect Hilbert space for 2+1d $U(1)$ pure gauge theory.

3.1 Symmetries

We work in the first presentation where the Gauss laws are modified (2.18) but the Hamiltonian is unmodified (2.20). The theory has a magnetic 0-form $U(1)_m^{(0)}$, with the charge operator

$$Q^{(0)} = \sum_p \left(N - \frac{\theta}{2\pi} \right)_p. \quad (3.1)$$

Eigenvalues of the local charge $N - \theta/2\pi$ are integers because of the third equation in the Gauss laws (2.18). Though N_p is not gauge invariant, $Q^{(0)}$ is since $\sum_p N_p = \sum_p (N + dA/2\pi)_p$, where each summand is gauge invariant. The charged point operator is the monopole e^{iB_p} , which is not involved in the Hamiltonian, implying $[Q^{(0)}, H] = 0$.

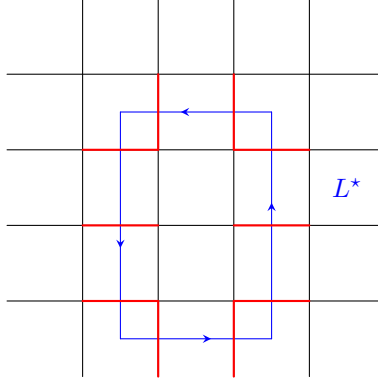


Figure 1: Symmetry defect for electric 1-from $U(1)_e^{(1)}$ symmetry.

The theory also has an electric 1-from $U(1)_e^{(1)}$ symmetry, with the charge operator

$$Q^{(1)} = \sum_{\ell^* \in L^*} \left(E + \frac{\delta B}{2\pi} - \frac{\lambda}{2\pi} \right)_\ell, \quad (3.2)$$

where L^* is the closed loop in the dual lattice, as in Figure 1. Eigenvalues of the local charge are integers because of the second equation in the Gauss laws (2.18). $Q^{(1)}$ is topological because of $[Q^{(1)}, H] = 0$ (in temporal direction) and $\delta(E - \delta B/2\pi - \lambda/2\pi) = 0$ (in spatial directions), which is also a part of the Gauss laws. The charged line operator is the Wilson line $\exp(i \sum_{\ell \in \gamma} A_\ell)$.

Even though $[Q^{(0)}, Q^{(1)}] = 0$, the local charges do not commute, as a manifestation the mixed anomaly similar to [5].

The continuous counterparts of the charge operators are

$$\mathcal{Q}^{(0)} = \int_\Sigma \frac{f}{2\pi}, \quad \text{and} \quad \mathcal{Q}^{(1)} = \int_L \frac{i}{e^2} \star f. \quad (3.3)$$

Strictly speaking, we have not derived the interpretation of the modified Gauss law, so the operator in the continuum is valid away from the spacetime region where modifications are made.

3.2 Topological defects

If the defect is topological, inserting such symmetry defects has been studied extensively in [1–4, 14, 32–37].

We start by inserting a $U(1)_e^{(1)}$ defect at plaquette p_0 . This is achieved by conjugating the

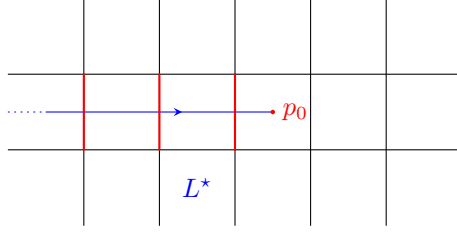


Figure 2: Half-line operator $U^{(1)}(\alpha, p_0)$ to insert a $U(1)_e^{(1)}$ defect at plaquette p_0 .

system (including both the Gauss laws and the Hamiltonian) by a truncated $U(1)_e^{(1)}$ symmetry operator $e^{i\alpha Q^{(1)}}$ where $Q^{(1)}$ is (3.2) but defined on truncated line L^* with a boundary at $\partial L^* = p_0$ as in Figure 2,

$$\begin{aligned} U^{(1)}(\alpha, p_0) &= \exp \left[i\alpha \sum_{L^*} \left(E - \frac{\delta B}{2\pi} - \frac{\lambda}{2\pi} \right)_\ell \right] \\ &= \exp \left[i\alpha \sum_{L^*} \left(E - \frac{\lambda}{2\pi} \right)_\ell \right] \exp \left(i \frac{\alpha}{2\pi} B_{p_0} \right). \end{aligned} \quad (3.4)$$

Conjugation by $U^{(1)}(\alpha, p_0)$ shifts $A_\ell \rightarrow A_\ell + \alpha$ for those $\ell^* \in L^*$ (red links in Figure 2), and shifts $N_{p_0} \rightarrow N_{p_0} - \alpha/2\pi$. Thus, the Hamiltonian H (2.20) and the first two Gauss law operators (2.18) are invariant, while the last Gauss law operator gets a phase $e^{-i\alpha}$ at p_0 . This means that inserting a $U(1)_e^{(1)}$ defect at plaquette p_0 changes the constraint from $G(n, \alpha, \tilde{n}) = \chi(n, \alpha, \tilde{n})$ to $G(n, \alpha, \tilde{n}) = \chi(n, \alpha, \tilde{n})e^{i\tilde{n}p_0\alpha}$, or equivalently

$$(\lambda, \theta) \rightarrow (\lambda, \theta + \alpha 1_{p_0}), \quad (3.5)$$

where 1_{p_0} is a 2-cochain which is 1 on plaquette p_0 and 0 elsewhere.

Note that this is a transitive action on the $C^2(\Lambda, \mathbb{R}/2\pi\mathbb{Z})$ factor of the moduli space $\mathcal{M}$ (2.19). Thus, the non-zero θ in the modified Gauss laws corresponds to the insertion of topological $U(1)_e^{(1)}$ line defects. If we shift to the second presentation in Table 1, this agrees with (4.27) in [38].

Now consider a $U(1)_m^{(0)}$ topological defect supported on a closed surface Σ with parameter $\alpha \in \mathbb{R}/2\pi\mathbb{Z}$. In the spacetime picture, the surface Σ intersects the equal time slice on a closed cycle L_0 , as in Figure 3. On the lattice, inserting this defect on L_0 is achieved by conjugating the system by a truncated $U(1)_m^{(0)}$ defect $e^{i\alpha Q^{(0)}}$ where $Q^{(0)}$ is (3.1) but defined on truncated

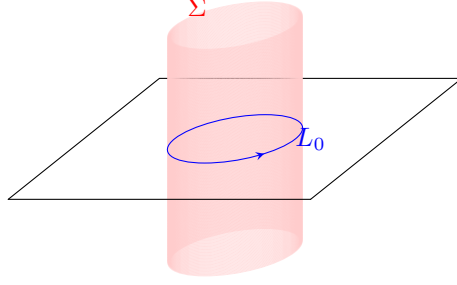


Figure 3: The spacetime picture for the $U(1)_m^{(0)}$ surface defect intersects with the equal time slice on a closed cycle L .

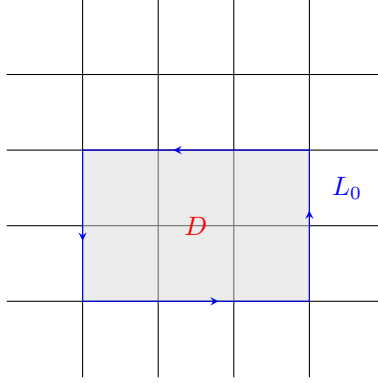


Figure 4: Inserting $U(1)_m^{(0)}$ defect along L_0 on the lattice.

surface D with a boundary at $\partial D = L_0$ as in Figure 4,

$$U^{(0)}(\alpha, L_0) = \exp \left(i\alpha \sum_{p \in D} \left(N - \frac{\theta}{2\pi} \right)_p \right). \quad (3.6)$$

Conjugating by $U^{(0)}(\alpha, L_0)$ shifts $B_p \rightarrow B_p + \alpha$ for those $p \in D$. Again, the Hamiltonian H (2.20), the first and the last Gauss law operators (2.18) are invariant, while the second Gauss law operator gets a phase $e^{i\alpha}$ at those $\ell \in L_0$. This means that inserting a $U(1)_e^{(1)}$ defect at L_0 changes the modified Gauss laws

$$(\lambda, \theta) \rightarrow (\lambda - \alpha 1_{L_0}, \theta). \quad (3.7)$$

Here 1_L is an integer 1-cochain which is $+1$ on the links of L with positive orientation, -1 on links taken the negative orientation, and 0 elsewhere.

Note that the 2π periodicity of α is compatible with the identification of λ in (2.19). Thus, (3.7) is a well-defined action on the moduli space $\mathcal{M}$.

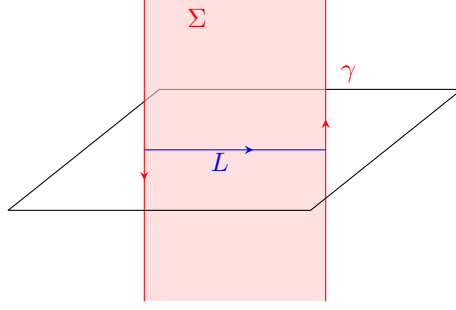


Figure 5: The spacetime picture for the defect $\mathcal{U}_x(\Sigma)$, where surface Σ with boundary γ intersects with the equal time slice on a segment L .

We also note that the modified Gauss law in 1+1d theory with $U(1)$ symmetry was discussed in [1, 5], where the modification is identified with twisting the boundary condition by $U(1)$ symmetry. This is a lower dimensional and lower symmetry-form analogue of what we discussed above.

3.3 Monodromy defects

However, the action (3.7) on the first factor of moduli space $\mathcal{M}$ is not transitive, since $\delta\lambda$ is invariant under such shift. This means starting with λ_0 , say the trivial one $\lambda_0 = 0$, we can only reach other value of λ with $\delta\lambda = \delta\lambda_0 = 0$ by inserting topological defect for $U(1)_m^{(0)}$ symmetry, and cannot reach the whole space of λ , i.e. the first factor of $\mathcal{M}$ in (2.19).

We start with the continuum and go beyond the realm of topological defects a little. Consider defect

$$\mathcal{U}_x(\Sigma) = \exp \left(ix \int_{\Sigma} \frac{F}{2\pi} \right), \quad (3.8)$$

where Σ is taken to be a surface with non-empty boundary γ , as in Figure 5. Now $\int_{\Sigma} F$ is not quantized on open surface, and hence $x \in \mathbb{R}$. The operator $\mathcal{U}_x(\Sigma)$ is invariant under infinitesimal deformation of the bulk of Σ , as long as $\gamma = \partial\Sigma$ is unchanged. In short, (3.8) is a truncated $U(1)_m^{(0)}$ surface defect.

Equivalent, $\mathcal{U}_x(\Sigma)$ is the monodromy defect of the monopole operator [38–42]. If we insert γ at $r = 0$ in space and along the time direction, denoting M_n for the charge n monopole operator, the boundary condition in the angular direction is

$$M_n(t, r, \theta + 2\pi) = e^{inx} M_n(t, r, \theta), \quad (3.9)$$

since the monopole gets acted by the surface operator once it pass through the defect. The

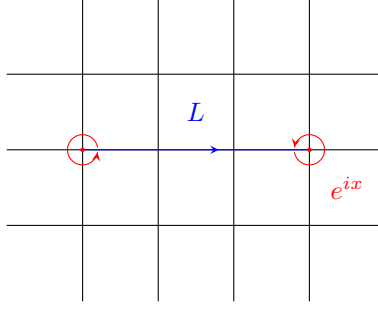


Figure 6: Inserting defect $\mathcal{U}_x(\Sigma)$ along L on the lattice. The monodromy is introduced on the ends of the segment L .

surface Σ is the branch cut for monodromy, hence topological. However, the boundary of the branch cut is non-topological.⁶

Let Σ intersects with the spatial lattice on links L and γ intersects on sites s , then L is a segment with ends $s = \partial L$, as in Figure 6. From the section 3.2, we conclude that inserting the defect (3.8) on the lattice, corresponds to shifting

$$\lambda_{\ell_*} \rightarrow \lambda_{\ell_*} - x \quad \text{for } \ell_* \in L. \quad (3.10)$$

This is to be contrasted with (3.7), where L are closed loops and $\alpha \in \mathbb{R}/2\pi\mathbb{Z}$. While in (3.10), L is an open segment and $x \in \mathbb{R}$.

One may ask whether the shift (3.10) can be achieved by conjugating a unitary operator, similar to (3.6). To realize such a shift in the modified Gauss law, one can conjugate it by the unitary

$$U^{(0)}(x, L) = \exp \left(-i \frac{x}{2\pi} \sum_{\ell \in L} A_\ell \right). \quad (3.11)$$

However, due to the dependence of A_ℓ , the unitary does not commute with the Hamiltonian. Hence the shift (3.10) is not achieved by merely conjugating a unitary operator to the system (including the Gauss law and the Hamiltonian).

Note that the action (3.10) on the space of λ is transitive. Start the trivial $\lambda = 0$, we can always insert defect $\mathcal{U}_{\lambda_\ell}$ on each link to obtain any nontrivial λ . The action (3.10) and (3.5) together is a transitive action on the moduli space $\mathcal{M}$. Hence we have understood that the physical Hilbert space $\mathcal{H}_\chi$, determined by the modified Gauss law $G = \chi$, is the defect Hilbert space with $U(1)_e^{(1)}$ topological line defects, and monodromy (or twist) defects for $U(1)_m^{(0)}$. Shifting between presentations in Table 1 do not change the result.

⁶Since $U(1)_m^{(0)}$ acts faithfully in the Maxwell theory, its topological surface operator can not be cut open topologically [43].

4 Magnetic translation

In this section, we study the physical Hilbert space $\mathcal{H}_\chi$ by looking at operators on $\mathcal{H}_\chi$, in particular the lattice translation operators. We show that for nontrivial χ , the lattice translations do not commute and satisfy the symmetry algebra for magnetic translations [44, 45], being a manifestation of the insertion monodromy defects concluded in Section 3.

Recall that (physical) operators of theory are those operators acting on the physical Hilbert space $\mathcal{H}_\chi$, i.e. $\mathcal{O} \in \text{End}(\mathcal{H}_\chi)$. Usually we start with an operator $\tilde{\mathcal{O}}$ on $\mathcal{H}$ and restrict it to the subspace $\mathcal{H}_\chi$, denoted by $\tilde{\mathcal{O}}|_{\mathcal{H}_\chi}$. We say $\tilde{\mathcal{O}}$ is physical if

$$G(n, \alpha) \tilde{\mathcal{O}} G^{-1}(n, \alpha)|_{\mathcal{H}_\chi} = \tilde{\mathcal{O}}|_{\mathcal{H}_\chi}, \quad (4.1)$$

i.e. $G\tilde{\mathcal{O}}G^{-1}$ and $\tilde{\mathcal{O}}$ acts on states in $\mathcal{H}_\chi$ in the same way. Equivalently, since $G^{-1} = \chi^{-1}$ on $\mathcal{H}_\chi$, (4.1) can also be recast as

$$G(n, \alpha) \tilde{\mathcal{O}}|_{\mathcal{H}_\chi} = \chi(n, \alpha) \tilde{\mathcal{O}}|_{\mathcal{H}_\chi}. \quad (4.2)$$

Motivated by the results in the last section, we would like to derive the lattice translation operator on $\mathcal{H}_\chi$. We start with the translation on $\mathcal{H}$, which acts on any local operator by

$$T_x \mathcal{O}_c T_x^{-1} = \mathcal{O}_{c+\hat{x}}, \quad T_y \mathcal{O}_c T_y^{-1} = \mathcal{O}_{c+\hat{y}}, \quad (4.3)$$

where c refers to a site, link, or plaquette, and $\hat{x}$, $\hat{y}$ are unite vectors on the square lattice Λ . The conventions for labeling neighboring sites etc. are listed in Appendix A.

Let χ correspond to $(\lambda, \theta) \in \mathcal{M}$. Generally, λ and θ take different values on links and plaquettes, making that $T_x G T_x^{-1}|_{\mathcal{H}_\chi} \neq \chi$. By (4.2), T_x is not a physical operator. However, we propose that a modified translation can be defined as follows,

$$\begin{aligned} T_1 &= T_x \exp \left(i \sum_p \lambda_{p+\frac{\hat{x}}{2}} N_p \right) \exp \left(-i \sum_{\ell \parallel \hat{x}} (\delta\lambda)_{\ell+\frac{\hat{x}}{2}} \frac{A_\ell}{2\pi} \right) \exp \left(-i \sum_p \theta_p \left(E + \frac{\delta B}{2\pi} \right)_{p-\frac{\hat{x}}{2}} \right) \\ T_2 &= T_y \exp \left(-i \sum_p \lambda_{p+\frac{\hat{y}}{2}} N_p \right) \exp \left(-i \sum_{\ell \parallel \hat{y}} (\delta\lambda)_{\ell+\frac{\hat{y}}{2}} \frac{A_\ell}{2\pi} \right) \exp \left(-i \sum_p \theta_p \left(E + \frac{\delta B}{2\pi} \right)_{p-\frac{\hat{y}}{2}} \right) \end{aligned} \quad (4.4)$$

satisfying (4.2), making it the translation on the physical Hilbert space $\mathcal{H}_\chi$. Here by $\ell \parallel \hat{x}$ we only sum over those links parallel to x direction, similar for $\ell \parallel \hat{y}$. Again, we work in the first presentation in Table 1.

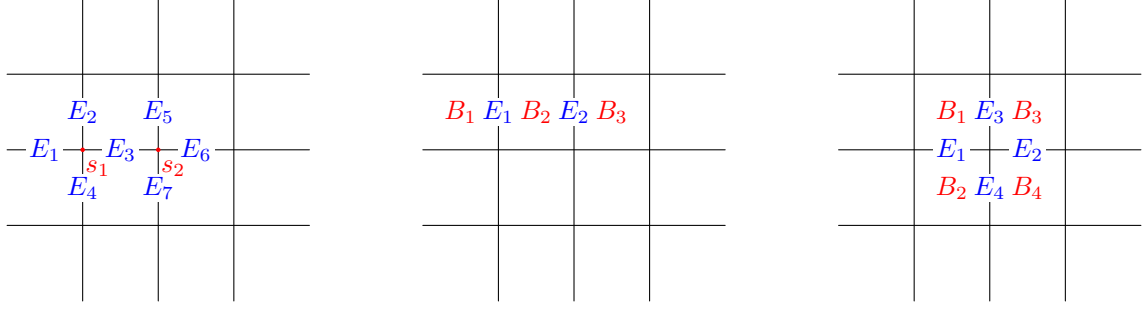


Figure 7: Conjugating the local terms in Gauss law operator with the modified translation operator T_1 . The three panels shows the label for neighboring links and plaquettes.

Proof. In the following, we give the derivation for (4.4).

Since T_i is invertible, we prove (4.2) by showing that $T_i G T_i^{-1}|_{\mathcal{H}_\chi} = \chi$. From (2.18), we only need to conjugate T_i on each local term in the Gauss law operator G . Take T_1 as an example. For any site, say s_1 in Figure 7 (left), only the terms involving horizontal neighboring links, i.e. A_1 and A_3 , and T_x conjugate non-trivially on $(\delta E)_{s_1}$,

$$\begin{aligned}
& T_x \exp\left(-i(\delta\lambda)_{s_1} \frac{A_1}{2\pi} - i(\delta\lambda)_{s_2} \frac{A_3}{2\pi}\right) (\delta E)_{s_1} \exp\left(i(\delta\lambda)_{s_1} \frac{A_1}{2\pi} + i(\delta\lambda)_{s_2} \frac{A_3}{2\pi}\right) T_x^{-1} \\
&= T_x (\delta E)_{s_1} T_x^{-1} + \frac{(\delta\lambda)_{s_1}}{2\pi} - \frac{(\delta\lambda)_{s_2}}{2\pi} \\
&= (\delta E)_{s_2} + \frac{(\delta\lambda)_{s_1}}{2\pi} - \frac{(\delta\lambda)_{s_2}}{2\pi}.
\end{aligned} \tag{4.5}$$

Restricting to $\mathcal{H}_\chi$, it equals $\frac{(\delta\lambda)_{s_1}}{2\pi}$, which means that δE is invariant when conjugated with T_1 . For a plaquette p , $T_1 \exp(2\pi i N_p) T_1^{-1}$ is calculated similarly, and $\exp(2\pi i N_p)$ is invariant restricted on $\mathcal{H}_\chi$.

For any vertical link, say the one on which is E_1 in Figure 7 (middle), only the terms involving neighboring plaquettes, i.e. N_1 and N_2 , and T_x conjugate non-trivially on $e^{i(2\pi E_1 + B_1 - B_2)}$,

$$\begin{aligned}
& T_x e^{i(\lambda_1 N_1 + \lambda_2 N_2)} e^{i(2\pi E_1 + B_1 - B_2)} e^{-i(\lambda_1 N_1 + \lambda_2 N_2)} T_x^{-1} \\
&= T_x e^{i(2\pi E_1 + B_1 - B_2)} T_x^{-1} e^{i(\lambda_1 - \lambda_2)} \\
&= e^{i(2\pi E_2 + B_2 - B_3)} e^{i(\lambda_1 - \lambda_2)}.
\end{aligned} \tag{4.6}$$

Restricting to $\mathcal{H}_\chi$, it equals $e^{i\lambda_1}$, which means that $e^{i(2\pi E_1 + B_1 - B_2)}$ is invariant when conjugated with T_1 .

Finally, for any horizontal link, say the one on which is E_1 in Figure 7 (right), the terms

involving N_1 , N_2 , A_1 , and T_x conjugate non-trivially on $e^{i(2\pi E_1+B_1-B_2)}$,

$$\begin{aligned}
& T_x e^{i(\lambda_1 N_1 + \lambda_2 N_2)} \exp\left(-i(\delta\lambda)_{s_1} \frac{A_1}{2\pi}\right) e^{i(2\pi E_1+B_1-B_2)} \exp\left(i(\delta\lambda)_{s_1} \frac{A_1}{2\pi}\right) e^{-i(\lambda_1 N_1 + \lambda_2 N_2)} T_x^{-1} \\
&= T_x e^{i(2\pi E_1+B_1-B_2)} T_x^{-1} e^{i(\lambda_1-\lambda_2)} \\
&= e^{i(2\pi E_2+B_3-B_4)} e^{i(\lambda_3-\lambda_4+(\delta\lambda)_{s_1})}.
\end{aligned} \tag{4.7}$$

Restricting to $\mathcal{H}_\chi$, it equals $e^{i(\lambda_2+\lambda_3-\lambda_4+(\delta\lambda)_{s_1})} = e^{i\lambda_1}$, which means that $e^{i(2\pi E_1+B_1-B_2)}$ is invariant when conjugated with T_1 . $\square$

In conclusion, we have proven that $T_1 G T_1^{-1}|_{\mathcal{H}_\chi} = \chi$, which implies that T_1 is an physical operator on $\mathcal{H}_\chi$. Since it is obtained by modifying the translation on $\mathcal{H}$, we conclude that T_1 is the x direction translation operator in $\mathcal{H}_\chi$. The proof for T_2 is similar.

Note that the translations in two directions do not commute. Motivated by the conclusions in Section 3, i.e. monodromy defect insertions, we calculate

$$T_1 T_2 T_1^{-1} T_2^{-1} = \exp\left[-i \sum_s (\delta\lambda)_s \left(N + \frac{dA}{2\pi}\right)_{s+\frac{\hat{x}}{2}+\frac{\hat{y}}{2}}\right]. \tag{4.8}$$

Since $(N + \frac{dA}{2\pi})_p$ is the gauge invariant local monopole density operator, the magnetic translation algebra (4.8) is a manifestation of the $e^{i(\delta\lambda)_s}$ monodromy at each site.

Note that (4.8) do not depend on the part of (λ, θ) which corresponds to the insertion of *topological* defects, i.e those discussed in Section 3.2, but only depend on the insertion of non-topological monodromy defects. This agrees with the expectation that inserting topological defects alone do not give raise to magnetic translations.

Also, for the Hamiltonian H (2.20) with $e \neq 0$, the translation operators T_i (4.4) do not commute with the Hamiltonian H (2.20) if $\delta\lambda \neq 0$, meaning that the (magnetic) translation symmetry is *explicitly* broken. Note that the Gauss law with $\delta\lambda \neq 0$ is obtained by inserting monodromy defects. This agrees with the fact that the monodromy defect is not topological, which breaks the translation symmetry.

5 Duality to compact boson

In the continuum, 2+1d free Maxwell theory is dual to free compact boson at radius $R = \frac{e}{2\pi}$,

$$\frac{1}{2e^2} f \wedge \star f \quad \longleftrightarrow \quad \frac{e^2}{8\pi^2} d\phi \wedge \star d\phi. \tag{5.1}$$

In this section, we demonstrate this exact duality in the Hamiltonian lattice model, and compare the Gauss laws on both sides.

The lattice duality was discussed in [16]. Again we start with the square lattice with periodic boundary conditions, and put a pair of real-valued conjugate variables on each site and link,

$$[\phi_s, p_{s'}] = i\delta_{ss'}, \quad [\tilde{\phi}_\ell, w_{\ell'}] = i\delta_{\ell\ell'}. \quad (5.2)$$

The 2π shift of ϕ_s is gauged by w_ℓ , hence the following $\mathbb{Z}$ gauge transformation

$$\phi \rightarrow \phi + 2\pi n, \quad w \rightarrow w - dn \quad (5.3)$$

for $n \in C^0(\Lambda, \mathbb{Z})$. Except the 2π periodic⁷, $\tilde{\phi}$ has a more secret gauge redundancy by sifting $\delta\alpha$,

$$\tilde{\phi} \rightarrow \tilde{\phi} + 2\pi\tilde{n} + \delta\alpha, \quad (5.4)$$

for $\tilde{n} \in C^1(\Lambda, \mathbb{Z})$ and $\alpha \in C^2(\Lambda, \mathbb{R})$. The Gauss law operator can be written down similar to (2.9),

$$G = \exp \left[i \sum_s n_s \left(2\pi p_s + \delta\tilde{\phi}_s \right) \right] \exp \left[i \sum_p \alpha_p (dw)_p + i \sum_\ell 2\pi\tilde{n}_\ell w_\ell \right] \quad (5.5)$$

and we first do the unmodified constraints for simplicity,

$$e^{i(2\pi p + \delta\tilde{\phi})} = 1, \quad dw = 0, \quad e^{2\pi i w} = 1. \quad (5.6)$$

The Hamiltonian for the compact boson theory is

$$H = \sum_s \frac{1}{2R^2} p_s^2 + \sum_\ell \frac{R^2}{2} (d\phi + 2\pi w)_\ell^2. \quad (5.7)$$

The charge operators for 0-from $U(1)_{\text{shift}}^{(0)}$ and 1-from $U(1)_{\text{winding}}^{(1)}$ symmetries are

$$Q^{(0)} = \sum_s \left(p + \frac{\delta\tilde{\phi}}{2\pi} \right)_s, \quad Q^{(1)} = \sum_\ell w_\ell. \quad (5.8)$$

The reasons for these charge operators being topological, and the local charges being integer-

⁷or the eigenvalues of w_ℓ being integers

valued are similarly to the discussions in Section 3.1.

Now we introduce a new set of variables to describe this theory, defined by

$$\begin{aligned} A_\ell &= \tilde{\phi}_\ell, & E_\ell &= w_\ell + \left(\frac{d\phi}{2\pi} \right)_\ell, \\ B_s &= \phi_s, & N_s &= p_s + \left(\frac{\delta\tilde{\phi}}{2\pi} \right)_s. \end{aligned} \tag{5.9}$$

If we relabel variables by the dual lattice⁸, it can be checked that these variables satisfy the commutation relations (2.1) (2.2) in the abelian gauge theory, the Hamiltonian of the boson theory (5.7) takes exactly the same form as (2.20) with $e = 2\pi R$, so are the 0-from and 1-from charge operators. In particular, we focus on the constraints

$$\begin{aligned} e^{i(2\pi p + \delta\tilde{\phi})} = 1 &\longleftrightarrow e^{2\pi i N} = 1 \\ dw = 0 &\longleftrightarrow dE = 0 \\ e^{2\pi i w} = 1 &\longleftrightarrow e^{i(2\pi E + dB)} = 1, \end{aligned} \tag{5.10}$$

which recovers the unmodified constraints in the abelian gauge theory (2.11).

We make two comments here. Firstly, the compact boson in the continuum is not a gauge theory in an obvious way, but the corresponding lattice model has constraints, that are matched with the lattice gauge theory. Secondly, in the compact boson presentation, the constraint $e^{2\pi i w} = 1$ which fixes the eigenvalues of w_ℓ to be integers, gets mapped to the Gauss law in the U(1) gauge theory presentation, as shown in (5.10). So the gauge transformation associated with the gauge group is not an unambiguous notion, and we suggest just regard them as constraints in the Hamiltonian lattice model.

Of course, we can work out all possible modified Gauss laws for the Gauss law operator (5.5) in the compact boson theory. The calculation and the result, i.e. the moduli space $\mathcal{M}$, are basically the same as Section 2.2, matching with the U(1) gauge theory presentation.

6 Discussions and future directions

One generalization is that we couple the U(1) gauge field to matter degrees of freedom, making it an interacting theory. A subtlety in the lattice is that if the eigenvalues of the local charge operator of the matter is finite, such as $\rho_s = \sigma_s^z$ for a qubit whose eigenvalues are ± 1 , the Gauss law with some choice of χ could lead to non-extensive or empty physical

⁸namely exchange sites with plaquettes, d with δ , etc..

Hilbert space.⁹ Hence the physical interpretation of the Gauss law is more subtle.

If we couple to a complex scalar field where the local Hilbert space is infinite dimensional, for example considering the abelian-Higgs model, then such problem does not exist and we can also think about the modified Gauss laws. Then part of the Gauss law operator changes

$$\exp\left(i\sum_s\alpha_s(\delta E)_s\right)\rightarrow\exp\left(i\sum_s\alpha_s(\delta E+\rho)_s\right). \quad (6.1)$$

Some consistency conditions used in Section 2.2 no longer hold, hence the moduli space we discussed does not directly apply to the abelian-Higgs model. Despite this, we can still discuss the interpretation of the modification (2.18) or (2.22).

Note that introducing the scalar does not affect the third modified Gauss law in (2.18), i.e. $\exp(2\pi iN - i\theta) = 1$. In the abelian-Higgs model this cannot correspond to the insertion of $U(1)_e^{(1)}$ line defect since the $U(1)_e^{(1)}$ symmetry is explicitly broken by the charge 1 complex scalar, but the insertion of vortex loop defect:

$$\lim_{r\rightarrow 0}\exp\left(i\int_{C_r}a\right)=e^{i\theta}. \quad (6.2)$$

If we shift to the modified Hamiltonian presentation in Tabel 1, this has been discussed in [38]. Similar phenomenon also occurs in finite lattice gauge theories, see Appendix C.¹⁰

On the other hand, the magnetic $U(1)_m^{(0)}$ symmetry still exists. So we claim that the modification of λ still corresponds to the insertion of monodromy defect in the interacting theory.

Other future directions include analyzing the modified Gauss laws for nonabelian gauge theories on the lattice [23], possible realization of the anomalous translation [48, 49] on the lattice, and modifying the non-invertible Gauss law [50].

Acknowledgments

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⁹This subtlety has already been noticed in Kogut and Susskind [46, 47] and recently in [14].

¹⁰In the finite lattice gauge theory, adding matter degrees of freedom can be achieved by only modifying the Hamiltonian, without introducing degrees of freedom on sites. See Appendix C for details.



Figure 8: Left: the definition of operators ∂ and d . Right: the definition of operators $\hat{\delta}$ and δ .

A Conventions on (co)chains on square lattice

We consider 2d square lattice Λ with periodic boundary conditions in both direction, where $s \in C_0(\Lambda, \mathbb{Z})$ denotes sites, $\ell \in C_1(\Lambda, \mathbb{Z})$ denotes links and $p \in C_1(\Lambda, \mathbb{Z})$ denotes plaquettes. The unit vectors in positive x and y directions are denoted as $\hat{x}$ and $\hat{y}$. The links are oriented in the positive x or y direction. The direction reversed for ℓ is denoted as $-\ell$.

The boundary operator is defined as $\partial : C_k(\Lambda, \mathbb{Z}) \rightarrow C_{k-1}(\Lambda, \mathbb{Z})$. In particular,

$$\partial \ell = s_2 - s_1, \quad \text{and} \quad \partial p = \ell_1 + \ell_2 - \ell_3 - \ell_4, \quad (\text{A.1})$$

in Figure 8.

Let A be an abelian group, like $\mathbb{R}$ or $\mathbb{Z}$. The coboundary operator is defined as $d : C^k(\Lambda, A) \rightarrow C^{k+1}(\Lambda, A)$, by

$$(da)_c = a_{\partial c}. \quad (\text{A.2})$$

In particular

$$(d\phi)_\ell = \phi_{s_2} - \phi_{s_1}, \quad \text{and} \quad (da)_p = a_{l_1} + a_{l_2} - a_{l_3} - a_{l_4}, \quad (\text{A.3})$$

for $\phi \in C^0(\Lambda, A)$ and $a \in C^1(\Lambda, A)$ in Figure 8.

We also use the operator $\hat{\delta} : C_k(\Lambda, \mathbb{Z}) \rightarrow C_{k+1}(\Lambda, \mathbb{Z})$ defined by

$$\hat{\delta} c = \sum_{f \text{ such that } c \subset \partial f} f. \quad (\text{A.4})$$

As in Figure 8,

$$\hat{\delta} s = \ell_1 + \ell_2 - \ell_3 - \ell_4, \quad \text{and}, \quad \hat{\delta} \ell = p_1 - p_2. \quad (\text{A.5})$$

The corresponding operator acting on cochains $\delta : C^k(\Lambda, A) \rightarrow C^{k-1}(\Lambda, A)$, is defined by

$$(\delta a)_c = a_{\hat{\delta}c}. \quad (\text{A.6})$$

As in Figure 8,

$$(\delta a)_s = a_{\ell_1} + a_{\ell_2} - a_{\ell_3} - a_{\ell_4}, \quad \text{and} \quad (\delta b)_\ell = b_{p_1} - b_{p_2} \quad (\text{A.7})$$

Actually, δ has an equivalent definition. View $C^k(\Lambda, \mathbb{R})$ as a real vector space, with the inner product

$$\langle a, b \rangle = \sum_{c \in C_k(\Lambda)} a_c b_c. \quad (\text{A.8})$$

Since d is a linear map, we define δ as the adjoint of d with respect to such inner product, i.e.

$$\langle a, db \rangle = \langle \delta a, b \rangle. \quad (\text{A.9})$$

It can be shown by direct calculation that this definition is equivalent to the geometric definition (A.6). (A.9) can be thought of as the lattice version of integral by part and is used in the paper.

We also use the following notation for refer to neighboring links or plaquettes in Section 4.

$$(\text{A.10})$$

B Proof of (2.16): constructing λ from $\chi(n, \alpha, 0)$

We prove (2.16) in the appendix, i.e. there exist $\lambda \in C^1(\Lambda, \mathbb{R})$ such that

$$q = \frac{\delta\lambda}{2\pi}, \quad \text{and} \quad \varphi = \lambda \mod 2\pi. \quad (\text{B.1})$$

To start, we need to prove a lemma which is used in the following.

Lemma 1. *Let Λ be a square lattice with periodic boundary conditions in both directions. For any $m \in C^0(\Lambda, \mathbb{Z})$ satisfying $\sum_s m_s = 0$, there exists $j \in C^1(\Lambda, \mathbb{Z})$, such that $m = \delta j$.*

Proof. We provide an explicit construction of j from m .

For every site $s \neq (0, 0)$, we pick a nearest-neighbor path γ_s from the origin $(0, 0)$ to s . For example, if $s = (a, b)$, take the path that first move horizontally from $(0, 0)$ to $(a, 0)$, than vertically from $(a, 0)$ to (a, b) .

Define j by putting m_s along the path γ_s , for every $s \neq (0, 0)$,

$$j = \sum_{s \neq (0,0)} m_s 1_{\gamma_s}. \quad (\text{B.2})$$

Here 1_{γ_s} is an integer 1-cochain which is $+1$ on the links of γ_s with positive orientation, -1 on edges taken the negative orientation, and 0 elsewhere.

Note that

$$\delta 1_{\gamma_s} = 1_s - 1_{(0,0)}, \quad (\text{B.3})$$

where 1_s is an integer 0-cochain which is 1 on site s and 0 elsewhere. So

$$\begin{aligned} \delta j &= \sum_{s \neq (0,0)} m_s (1_s - 1_{(0,0)}) \\ &= \sum_{s \neq (0,0)} m_s 1_s - \left(\sum_{s \neq (0,0)} m_s \right) 1_{(0,0)} \\ &= \sum_{s \neq (0,0)} m_s 1_s + m_{(0,0)} 1_{(0,0)} \\ &= \sum_s m_s 1_s \equiv m, \end{aligned} \quad (\text{B.4})$$

where the third equality follows from $\sum_s m_s = 0$. Thus, j defined in (B.2) is the one stated

in the Lemma 1.

□

Now we can prove (2.16). As mentioned in the remarks below (2.10), $\chi(dk, -2\pi k, 0) = G(dk, -2\pi k, 0) = 1$, for $k \in C^0(\Lambda, \mathbb{Z})$. The second equality follows from the definition of Gauss law operator G (2.9). Thus,

$$\begin{aligned} 1 &= \chi(dk, -2\pi k, 0) = \chi(dk, 0, 0)\chi(0, -2\pi k, 0) \\ &= \exp \left(i \sum_{\ell} (dk)_{\ell} \varphi_{\ell} - 2\pi i \sum_s k_s q_s \right) \\ &= \exp \left(i \sum_s k_s (\delta\varphi - 2\pi q)_s \right), \end{aligned} \tag{B.5}$$

where the second equality follows from χ being a homomorphism, and the last equality follows from the property (A.9). Since this holds for any $k_s \in \mathbb{Z}$,

$$(\delta\varphi)_s = 2\pi q_s \mod 2\pi. \tag{B.6}$$

Recall that by definition $\varphi \in C^1(\Lambda, \mathbb{R}/2\pi\mathbb{Z})$, we pick an arbitrary lift from $\mathbb{R}/2\pi\mathbb{Z}$ to $\mathbb{R}$, i.e. $\tilde{\varphi} \in C^1(\Lambda, \mathbb{R})$,

$$\tilde{\varphi}_{\ell} = \varphi_{\ell} \mod 2\pi. \tag{B.7}$$

Then, by (B.6) there exist integer $m_s \in \mathbb{Z}$ such that

$$\delta\tilde{\varphi} - 2\pi q = 2\pi m. \tag{B.8}$$

Note that this equality holds on $\mathbb{R}$ since $q_s \in \mathbb{R}$.

Consider a configuration of $\alpha \in C^0(\Lambda, \mathbb{R})$ such the α_s takes a constant value on every site, $\alpha_s = x$. Since $\sum_s (\delta E)_s = 0$ on closed lattice (periodic boundary condition),

$$\exp \left(ix \sum_s q_s \right) = \chi(0, \alpha, 0) = G(0, \alpha, 0) = \exp \left(ix \sum_s (\delta E)_s \right) = 1, \tag{B.9}$$

imply that $\sum_s q_s = 0$. Thus, from (B.8) $m \in C^0(\Lambda, \mathbb{Z})$ satisfies

$$\sum_s m_s = 0 \tag{B.10}$$

From Lemma 1, there exists $j \in C^1(\Lambda, \mathbb{Z})$ such that $m = \delta j$, as constructed in the proof. Thus, from (B.8),

$$2\pi q = \delta(\tilde{\varphi} - 2\pi j). \quad (\text{B.11})$$

We define

$$\lambda = \tilde{\varphi} - 2\pi j \in C^1(\Lambda, \mathbb{R}), \quad (\text{B.12})$$

hence (2.16) is proven.

C Compare with lattice finite gauge theories

In this appendix, we revisit the Gauss laws in $\mathbb{Z}_2$ lattice models in 2+1d lattice gauge theories, following [33, 4], and compare them with the Gauss laws in the U(1) lattice gauge theory.

We consider a square lattice, with spin- $\frac{1}{2}$ gauge field degrees of freedom, and the local Hilbert space is $\mathcal{H}_\ell = \mathbb{C}^2$, acted upon by σ_ℓ^a , $a = x, y, z$ and ℓ labels the link. Similar to the U(1) examples in the main text, we don't include matter degrees of freedom on sites.

The modified Gauss law is

$$\prod_{s \in \partial \ell} \sigma_\ell^x = (-1)^{\alpha_s}. \quad (\text{C.1})$$

A Hamiltonian commuting with (C.1) is

$$H = g_1 \sum_p \prod_{\ell \in \partial p} \sigma_\ell^z + g_2 \sum_\ell \sigma_\ell^x. \quad (\text{C.2})$$

This is the $\mathbb{Z}_2$ lattice gauge theory studied in [51–54].¹¹

The Hamiltonian (C.2), for any value of g_1, g_2 , has a $\mathbb{Z}_2^m$ 1-form symmetry, generated by the unitary line operator

$$U(L^*) = \prod_{\ell \in L^*} \sigma_\ell^x, \quad (\text{C.3})$$

where L^* is a closed loop in the dual lattice. When $g_2 = 0$, the Hamiltonian (C.2) has an additional $\mathbb{Z}_2^e$ 1-form symmetry, generated by

$$V(L) = \prod_{\ell \in L} \sigma_\ell^z \quad (\text{C.4})$$

¹¹These papers impose the Gauss law constraint energetically and hence allows an additional interaction $\sum_\ell \sigma_\ell^z$. In our work, we always enforce the Gauss law strictly.

where L is a closed loop in the square lattice.

The modified Gauss law (C.1) can be mapped to the ordinary Gauss law by conjugating the truncated V operator

$$V(L, s) = \prod_{\ell \in L, s \in \partial L} \sigma_{\ell}^z. \quad (\text{C.5})$$

When $g_2 = 0$, since V is a $\mathbb{Z}_2^e$ symmetry generator, conjugation by (C.5) amounts to inserting a $\mathbb{Z}_2^e$ 1-form symmetry defect. Hence the modified Gauss law can be obtained from the ordinary Gauss law by inserting a $\mathbb{Z}_2^e$ 1-form symmetry defect.

When $g_2 \neq 0$, since V is no longer a symmetry, conjugating (C.5) with (C.2) flips the sign of σ_{ℓ}^x along L , hence the above interpretation fails. To see the physical meaning of the modified Gauss law, let's consider the continuum limit. The Hamiltonian (C.2) flows to a real scalar ϕ coupling to a dynamical $\mathbb{Z}_2$ gauge field a . The Lagrangian is

$$(D_{\pi a} \phi)^2 + m^2 \phi^2 + u \phi^4 \quad (\text{C.6})$$

The σ_{ℓ}^x on each link is regarded as a Wilson line segment $(-1)^{a_{\ell}}$ of the $\mathbb{Z}_2$ gauge field. The ordinary Gauss law $\prod_{s \in \partial \ell} \sigma_{\ell}^x = 1$ ensures that the gauge field is flat $(-1)^{da} = 1$. Suppose the Gauss law is modified at one site s . It means that the flatness of the $\mathbb{Z}_2$ gauge field is violated at one plaquette $p = s^*$ in the dual lattice. Around that plaquette, the $\mathbb{Z}_2$ gauge field has $\mathbb{Z}_2$ -flux, $(da)_{s^*} = 1$, or equivalently, the $\mathbb{Z}_2$ Wilson loop has a non-trivial phase

$$\exp \left(i\pi \oint_{\gamma} a \right) = -1, \quad \text{link}(\gamma, s) = 1 \quad (\text{C.7})$$

This defect is the $\mathbb{Z}_2$ -vortex, as studied in [38]. In summary, we find that modification of the Gauss law (C.1) is equivalent to inserting $\mathbb{Z}_2$ -vortices.

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